Le Minh Ngo

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Education

Lehigh University Bethlehem, PA

B.S. in Computer Science and Business; Minor in Applied Mathematics *Aug. 2025 – May 2029*

- **Cumulative GPA:** 3.73/4.00
- **Relevant Coursework:** Data Structures and Programming, Software Engineering, Linear Algebra, Probability and Statistics, Financial Accounting, Macroeconomics

Experience

Lehigh University Bethlehem, PA

Undergraduate Researcher | Computational Simulation & Network Science Lab *Aug. 2026 – Present*

- First-authored a research article submitted to *Vaccine*; reconstructed a 1,650-country-month panel from 576 cholera outbreaks and 88 oral cholera vaccine requests across 25 African countries, with reproducible code and derived data released on Zenodo
- Implemented pooled logistic regression with a between–within decomposition, country-clustered confidence intervals, and a 99,999-replication wild cluster bootstrap; found stronger between-country than within-country associations (OR 1.56 vs. 1.03; contrast $p = 0.022$)
- Led an external-documentation audit of 22 reactive request-months with zero recorded activity in the prior year; located reports documenting cholera for 17, showing that zeros in the linked series did not imply absence of documented cholera

Software Engineer | Lehigh360 AI Advising Hub *Sep. 2026 – Present*

- Built an AI advising workflow using OpenAI structured outputs, validating program recommendations against the Lehigh360 catalog and persisting student-, thread-, and eligibility-aware results in PostgreSQL via Prisma
- Connected Chat and Discover in Next.js and TypeScript, surfacing personalized program cards, recommendation rationales, save/apply actions, and resumable conversations with per-student authorization
- Hardened recommendation quality through deduplication, factual-query filtering, and eligibility checks; validated the end-to-end workflow with database, type, lint, and production-build tests

ML/Data Engineer | STEM Visualization Impact Fellowship *Jan. 2026 – Present*

- Developing a SINDy-based equation-discovery module in Python and Bokeh for Lehigh's public STEM Visualization platform, enabling researchers to infer governing equations directly from raw data
- Building data-processing and statistical-modeling workflows to characterize and predict the behavior of linear and periodic dynamical systems
- Co-developing interactive chemistry simulations for ideal gas laws and recycling systems on a four-person team, projected to support 200–300 students per semester

Teaching Assistant | Department of Mathematics *Jan. 2026 – Present*

- Grade weekly work for MATH 205, a 300-student course, and coach students on eigenvalue decomposition and differential equations; helped eliminate the course's grading backlog for the first time in two years

Student Assistant | Office of International Students and Scholars *Aug. 2025 – Sep. 2026*

- Triaged academic, work-authorization, and immigration-policy inquiries from 50+ international students and coordinated logistics for 20+ campus-wide events annually

Technical Skills

Languages & Web: Python, Java, SQL, TypeScript, JavaScript, HTML/CSS
Libraries & Tools: Next.js, PostgreSQL, Prisma, OpenAI API, pandas, statsmodels, Bokeh, Git/GitHub
Methods: SINDy, Panel Data, Logistic Regression, Between–Within Decomposition, Cluster-Robust Inference, Wild Cluster Bootstrap, Observational Study Design